PTF ALL

Equip Cooling Exhaust Fan Does Not Shut Off During A Cargo Compartment Smoke Alert - Fault Isolation

CORRECTIVE ACTION SUMMARY

Replace Exhaust System Control Interrupt Relay [R949]
Replace Forward Fire Alarm Latch Relay [R945]
Replace Recirculate Fan Switch Bypass Relay [R644]
Replace Smoke Control Relay [R648]
Replace Autosmoke Isolation Diode [R1251]
Replace Cargo Fire Control Panel [P8-75]
Repair Wiring D12229-D12225
Repair Wiring D12225-D11866
Repair Wiring D11866-D11874
Repair Wiring D11874-Circuit Breaker C1444

A. Initial Evaluation

(1) Do this task:
Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807.

(a) If the system does not operate as specified in the AMM task, then do the Fault Isolation Procedure below.

(b) If the system operates as specified in the AMM task, then there was an intermittent fault.

B. Fault Isolation Procedure

(1) Replace the exhaust system control interrupt relay, R949.
(a) Do the AMM test that you did at the start of this task.
(b) If the system operates as specified in the AMM task, then you corrected the fault.
(c) If the system does not operate as specified in the AMM task, then continue.

(2) Replace the forward fire alarm latch relay, R945.
(a) Do the AMM test that you did at the start of this task.
(b) If the system operates as specified in the AMM task, then you corrected the fault.
(c) If the system does not operate as specified in the AMM task, then continue.

(3) Replace the recirculate fan switch bypass relay, R644.
(a) Do the AMM test that you did at the start of this task.
(b) If the system operates as specified in the AMM task, then you corrected the fault.
(c) If the system does not operate as specified in the AMM task, then continue.

(4) Replace the smoke control relay, R648.
(a) Do the AMM test that you did at the start of this task.
(b) If the system operates as specified in the AMM task, then you corrected the fault.
(c) If the system does not operate as specified in the AMM task, then continue.

(5) Do this check of the autosmoke isolation diode:
(a) Disconnect the wires from the power terminals of the autosmoke isolation diode, R1251 (WDM 21-43-21).
(b) Connect an ohmmeter to do this diode check in the forward direction:
1) Connect the positive probe of the meter to terminal A5 of the autosmoke isolation diode.
2) Connect the negative probe of the meter to terminal B5 of the autosmoke isolation diode.
3) There will be continuity between the pins if the forward direction is good.

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(c) Connect the ohmmeter to do this diode check in the reverse direction:
1) Connect the negative probe of the meter to terminal A5 of the autosmoke isolation diode.
2) Connect the positive probe of the meter to terminal B5 of the autosmoke isolation diode.
3) There will be no continuity between the pins if the reverse direction is good.

(d) If the diode check was not good in the forward direction or the reverse direction, then do these steps:
1) Replace the autosmoke isolation diode, R1251.
2) Do the AMM test that you did at the start of this task.
3) If the system operates as specified in the AMM task, then you corrected the fault.

(e) If the diode check was good in the forward and the reverse directions, then continue.

(f) Re-connect the wires to the power terminals of the autosmoke isolation diode, R1251.

(6) Replace the cargo fire control panel, P8-75. These are the tasks:
- Cargo Fire Control Panel Removal, AMM TASK 26-00-02-000-801,
- Cargo Fire Control Panel Installation, AMM TASK 26-00-02-400-801.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(7) Do this check of the wiring:

(a) Disconnect connector D12229 from the exhaust system control interrupt relay, R949.

(b) Disconnect connector D12225 from the forward fire alarm latch relay, R945.

(c) Do a wiring check between this pin of connector D12229 at the exhaust system control interrupt relay and this pin of connector D12225 at the forward fire alarm latch relay (WDM 21-27-21, WDM 21-43-21):

D12229		**D12225**
pin X1	.....	pin 8

(d) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D12229 to the exhaust system control interrupt relay, R949.
3) Re-connect connector D12225 to the forward fire alarm latch relay, R945.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

(f) Re-connect connector D12229 to the exhaust system control interrupt relay, R949.

(8) Do this check of the wiring:

(a) Disconnect connector D11866 from the recirculate fan switch bypass relay, R644.

(b) Do a wiring check between this pin of connector D12225 at the forward fire alarm latch relay and this pin of connector D11866 at the recirculate fan switch bypass relay (WDM 21-43-21):

D12225		**D11866**
pin 7	.....	pin 4

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D12225 to the forward fire alarm latch relay, R945.
3) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D12225 to the forward fire alarm latch relay, R945.

(9) Do this check of the wiring:

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(a) Disconnect connector D11874 from the smoke control relay, R648.

(b) Do a wiring check between this pin of connector D11866 at the recirculate fan switch bypass relay and this pin of connector D11874 at the smoke control relay (WDM 21-43-21):

D11866		**D11874**
pin 5	.....	pin B1

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.
3) Re-connect connector D11874 to the smoke control relay, R648.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.

(10) Do this check of the wiring:

(a) Open the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].

(b) Do a wiring check between this pin of connector D11874 at the smoke control relay and this terminal of circuit breaker C1444 (WDM 21-43-21):

D11874		**C1444**
pin B2	.....	term =L

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D11874 to the smoke control relay, R648.
3) Close the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

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